

ADEMP-Pre-Registration for Simulation Studies

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Template based on the paper:

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1 General Information

1.1 What is the title of the project?

Answer: Crowdsourcing methodological comparisons: A many-analysts simulation study to identify conditions under which results in convenience samples generalize to the target population.

1.2 Who are the current and future project contributors?

Answer: Mona Niethammer, Carolin Strobl, Anne-Laure Boulesteix, Eiko Fried, Tim Morris, Christina Sauer, Edouard Bonneville, Balazs Aczel, Suzanne Hoogeveen, Alexandra Sarafoglou, Sabine Hoffmann and ideally many more analysts.

1.3 Provide a description of the project.

Answer: We will conduct a many-analysts simulation study centred on an illustrative example designed to reveal how analyst decisions shape simulation outcomes. Within a shared set of constraints, analysts will define their own data-generating processes (DGPs) and construct convenience samples by imposing range restrictions on X and Y . The study's main objective is to quantify between-analyst variability in effect-size estimates when addressing an identical simulation problem. We evaluate five widely used effect-size measures in terms of relative and absolute bias, and power under different restriction scenarios.

1.4 Did any of the contributors already conduct related simulation studies on this specific question?

Answer: Yes, we conducted several preliminary simulation studies in a pilot-style manner to iteratively refine the materials provided to the analysts. The design of the materials and the formulation of our hypotheses were informed by prior literature on the topic (Bobko et al., 2001; Carretta & Ree, 2022; Hoogeveen, Sarafoglou, Aczel, et al., 2023; Hoogeveen, Sarafoglou, van Elk, & Wagenmakers, 2023; Sackett et al., 2007).

2 Aims

2.1 What is the aim of the simulation study?

Answer: The aim of this project is to empirically quantify how much results may vary when analysing simulated data, given the researcher degrees of freedom (RDF) inherent in methodological comparison studies. We focus on evaluating how well five different effect size estimates derived from convenience samples generalize to the full population. To this end, analysts will construct convenience samples by imposing range restrictions on X and Y , and we will examine the resulting relative and absolute bias of each estimator. In addition, we seek to understand which RDF influence the DGP. Analysts therefore specify their own DGP within a set of predefined constraints (see Section 3.1).

3 Data-Generating Process

3.1 How will the parameters for the data-generating process (DGP) be specified?

Answer: Each analyst defines their own DGP, subject to restrictions specified for the confirmatory stage (see “ADEMP confirmatory.pdf”).

3.2 What will be the different factors of the data-generating mechanism?

Answer: See “ADEMP confirmatory.pdf”.

3.3 If possible, provide specific factor values for the DGP as well as additional simulation settings.

Answer: Because this is a many-analysts simulation study, the specific numerical values used in the DGP are determined individually by each analyst. Thus, we do not prescribe concrete factor values. Instead, we define the structural components of the DGP and the dimensions along which each analyst must introduce systematic variation.

We require that the outcome is generated as $Y \sim N(0, \sigma^2)$, where σ^2 , the

intercept β_0 , and any additional regression coefficients remain fixed across all simulation conditions within each analyst's design. The predictor $\mathbf{X}$ may be generated as a continuous or a binary variable, depending on the analyst's conceptual approach. Analysts may alternatively generate both versions and submit them in separate files.

All other DGP components (e.g., sample size, distributional form of $\mathbf{X}$, error variance) may be freely chosen by the analyst but must remain constant within their simulation conditions and be documented in full to ensure reproducibility. This framework ensures that all analysts vary the same conceptual factors while retaining methodological flexibility in specifying their own DGPs.

The analysts are asked to define a `dgp()` function in *R*, which should return a list with two data frames: `convenience_x` and `convenience_y`, respectively. If `restrict = "none"`, i.e., when the argument "none" is passed to `restrict`, no restriction is applied. In this case, `convenience_x` and `convenience_y` should both contain the same unrestricted dataset.

Each dataset must include $\mathbf{X}, \mathbf{Y}$, and any additional covariates. Analysts have to specify the model formula (e.g., $\mathbf{Y} \sim \mathbf{X}$) and run a validation function to verify that the confirmatory constraints are satisfied and that the estimator for β_1 is unbiased. Otherwise, the analyst have to check for warnings before running the simulation. If the analysts have problems running the simulations they have the opportunity to contact us via mail.

3.4 If there is more than one factor: How will the factor levels be combined and how many simulation conditions will this create?

Answer: See "ADEMP confirmatory.pdf".

4 Estimands and Targets

4.1 What will be the estimands and/or targets of the simulation study?

Answer: See "ADEMP confirmatory.pdf".

5 Methods

5.1 How many and which methods will be included and which quantities will be extracted?

Answer: All analysts will use the same pre-specified R script to estimate the five effect sizes and run the simulations. For more details see “ADEMP confirmatory.pdf”.

Planned Meta-Regression Analyses. For each effect size measure and convenience-sample type, the outcome in the meta-regression models is the DGP-level mean relative bias. Let y_i denote the mean relative bias for observation i , where each observation corresponds to one condition within one analyst-specified DGP. Because Monte Carlo error is negligible with 10,000 replications, and because our goal is to characterize variability across DGPs rather than simulation uncertainty, all observations are equally weighted. Accordingly, we will fit unweighted multilevel meta-regression models of the form:

$$y_i = \beta_0 + \sum_{k=1}^K \beta_k x_{ki} + u_{d(i)} + v_{\alpha(i)} + e_i$$

where x_{ki} denotes the fixed-effect covariates, $u_{d(i)}$ the random intercept for DGP, $v_{\alpha(i)}$ the random intercept for analyst, and e_i the sampling error term. We assume:

$$u_{d(i)} \sim N(0, \tau_{DGP}^2), \quad v_{\alpha(i)} \sim N(0, \tau_{analyst}^2), \quad e_i \sim N(0, V_i), \quad (1)$$

with $V_i = 1$ for all observations. Thus, each DGP contributes equally to the estimation. The analyst-level random intercept is included only if at least one analyst contributes more than one DGP, for example by submitting separate designs for numeric and binary predictors. Otherwise the model reduces to:

$$y_i = \beta_0 + \sum_{k=1}^K \beta_k x_{ki} + u_{d(i)} + e_i.$$

For the unstandardized regression β_1 , the standardized regression coefficient β_1^{std} and Cohen’s d , we will fit random-effects meta-regression models using `metafor::rma.mv`. Two families of moderator models will be estimated. First, the **RDF-related models** examine whether variation in relative bias

is associated with RDF in the simulation design. The subjective RDF model includes analysts' categorical design choices as predictors:

$$y_i = \beta_0 + \beta_1 \text{RestrictionLevel}_i + \beta_2 \text{EffectSizeStrength}_i \\ + \beta_3 \text{ConvenienceSampleSize}_i + \beta_4 \text{PredictorType}_i + \beta_5 \text{DistributionType}_i \\ + \beta_6 \text{RestrictionMechanism}_i + \beta_7 \text{CovariatesIncluded}_i \\ + \beta_8 \text{CorrelatedCovariates}_i + u_{d(i)} + v_{a(i)} + e_i.$$

Here, *RestrictionLevel* refers to the subjective restriction categories weak, moderate, and strong; *EffectSizeStrength* refers to the subjective effect-size categories weak, moderate, and strong; and the remaining predictors indicate convenience-sample size, predictor type (numeric vs. binary), distribution type (parametric vs. non-parametric), restriction mechanism (deterministic vs. probabilistic), inclusion of covariates, and inclusion of correlated covariates. For Cohen's d , predictor type is omitted because this effect size is only defined for binary predictors.

The objective RDF model replaces the subjective design labels by quantitative characteristics of the implemented DGP:

$$y_i = \beta_0 + \beta_1 \text{VarianceReduction}_i + \beta_2 \text{TruePartialR}_i^2 \\ + \beta_3 \text{ConvenienceSampleSize}_i + \beta_4 \text{PredictorType}_i + \beta_5 \text{DistributionType}_i \\ + \beta_7 \text{CovariatesIncluded}_i + \beta_8 \text{CorrelatedCovariates}_i + u_{d(i)} + v_{a(i)} + e_i.$$

Again, predictor type is omitted for Cohen's d . In both RDF models, categorical predictors are dummy-coded with appropriate reference categories. Second, the **prior-belief model** examines whether analysts' pre-simulation plausibility ratings are associated with relative bias. Let Plausibility_i denote the relevant plausibility rating for the hypothesis under consideration. The model is:

$$y_i = \beta_0 + \beta_1 \cdot \text{Plausibility}_i + u_{d(i)} + v_{\alpha(i)} + e_i,$$

or, if no analyst contributes more than one DGP,

$$y_i = \beta_0 + \beta_1 \cdot \text{Plausibility}_i + u_{d(i)} + e_i.$$

Categories with insufficient observations, such as rarely selected RDF settings or plausibility levels, will be excluded from the corresponding models to ensure estimability. Because R^2 and Pearson's r are redundant with the standardized regression coefficient, no separate meta-regressions are estimated for these measures.

6 Performance Measures

6.1 Which performance measures will be used?

Answer: Performance will be evaluated using the *relative bias*, *absolute bias* and *statistical power*. For R^2 and r absolute rather than relative bias is shown because both R^2 and the correlation coefficient are already standardized measures, making absolute bias directly comparable across analysts and effect sizes measures without distortion due to scaling.

Let θ denote the true value of a given effect size measure and $\hat{\theta}_i$ the corresponding estimate in simulation i , for $i = 1, \dots, n_{\text{sim}}$.

Absolute and relative bias. The absolute bias of the estimator is

$$\widehat{\text{AbsBias}} = \frac{1}{n_{\text{sim}}} \sum_{i=1}^{n_{\text{sim}}} (\hat{\theta}_i - \theta).$$

For $\theta \neq 0$, we define the relative bias in simulation i as

$$b_i = \frac{\hat{\theta}_i - \theta}{\theta},$$

and summarize it across replications by the mean relative bias

$$\widehat{\text{RelBias}} = \frac{1}{n_{\text{sim}}} \sum_{i=1}^{n_{\text{sim}}} b_i.$$

These quantities will be reported as percentages ($\times 100$) for ease of interpretation.

The MCSE of the mean relative bias will be computed as

$$\text{MCSE}_{\widehat{\text{RelBias}}} = \frac{S_b}{\sqrt{n_{\text{sim}}}},$$

where S_b is the sample standard deviation of $(b_1, \dots, b_{n_{\text{sim}}})$. The MCSE for the absolute bias is calculated analogously by replacing *RelBias* with *AbsBias* and using the corresponding sample standard deviation.

Aggregation across estimands, conditions, and analysts. All performance measures will be first computed separately within each analyst’s DGP, for each effect size measure, each true effect-size level (β_1), each range-restriction level (none, weak, moderate, strong), and for both types of convenience samples (restricted on X vs. restricted on Y). Within a given DGP and condition, Monte Carlo replications will be summarized using the mean relative bias, its standard deviation, and the corresponding MCSE (with analogous summaries for all other performance measures).

In a second step, results will be aggregated across analysts by treating each DGP as one unit of analysis. Only valid submissions, as determined independently by two reviewers, will be included. Our analyses focus on the true partial R^2 as a continuous measure of effect-size strength, rather than on the pre-specified subjective effect-size categories (weak, moderate, strong). For each effect size measure, true partial R^2 , restriction level, and sample type, we will use locally estimated scatterplot smoothing (LOESS) (using `ggplot2::geom_smooth` with default settings and `span = 1`) and its 95% confidence band. Furthermore, as a quantitative measure for restriction we estimate variance reduction for each restriction level and sample (restricted on X or Y) as:

$$var_{red} = 1 - \frac{var_{restricted}}{var_{notrestricted}}.$$

For the pre-specified subjective effect-size categories (weak, moderate, strong), we will compute the unweighted mean and the empirical of the DGP-specific mean relative biases:

$$\bar{b}_{r,\beta} = \frac{1}{D} \sum_{d=1}^D \bar{b}_{d,r,\beta},$$

where D denotes the number of valid DGPs contributing to the condition. Uncertainty of the aggregated mean bias will be quantified using the 95% confidence interval based on the standard errors across DGPs:

$$CI_{0.95} = \bar{b}_{r,\beta} \pm 1.96 \frac{s_{r,\beta}}{\sqrt{D}}$$

where $s_{r,\beta}$ denotes the standard deviation of DGP-specific mean biases within the respective condition. Confirmatory evaluation will be based on those CIs, by comparing them to a $\pm 5\%$ equivalence region. Effect sizes will be classified as unbiased, biased, or inconclusive depending on whether the interval lay entirely within, entirely outside, or overlapped $[-0.05, 0.05]$.

In addition, we will report the empirical 2.5% and 97.5% quantiles of the DGP-specific mean biases to describe heterogeneity across analysts.

We will not aggregate across effect size measures; instead, results for all five effect size will be shown side by side using these summaries. Power will be not aggregated across DGPs; rather, all DGP-specific power estimates will be displayed jointly in a single plot.

6.2 How will Monte Carlo uncertainty of the estimated performance measures be calculated and reported?

Answer: Monte Carlo uncertainty will be quantified and reported at two levels: (1) within each analyst’s DGP and (2) across analysts.

Within-analyst MC uncertainty. See “ADEMP confirmatory.pdf”.

Across-analyst MC uncertainty. To summarize uncertainty across analysts, we will treat each DGP as one independent unit. For each combination of effect size measure, true effect size level, restriction level, and sample type, we will compute:

- the unweighted mean of the DGP-specific mean relative bias,
- a 95% confidence interval for this aggregated mean based on the standard error across DGPs, and
- the empirical 2.5% and 97.5% quantiles of the DGP-specific mean performance measures.

The 95% confidence interval reflects uncertainty in the estimated average performance across DGPs. In contrast, the empirical quantiles describe heterogeneity between analyst-specified DGPs and represent the spread of performance across different analytic designs. These quantiles are not interpreted as inferential confidence intervals.

For visualizations based on LOESS, we display pointwise 95% confidence bands around the smoothed mean trends across DGPs. These bands reflect uncertainty in the estimated smoothed average bias as a function of the true partial R^2 .

6.3 How many simulation repetitions will be used for each condition?

Answer: See “ADEMP confirmatory.pdf”.

6.4 How will missing values due to non-convergence or other reasons be handled?

Answer: Missing values and non-convergence are not expected. In cases of non-convergence, the respective analyst may be asked to submit a revised, convergent implementation. Missing values will be coded as NA and will not be imputed.

6.5 How do you plan on interpreting the performance measures? (optional)

Answer: For analyst’s outcome and interpretation across restriction levels and effect size measures see “ADEMP confirmatory.pdf”.

Meta-regression for RDF. The resulting regression coefficients quantify the expected change in relative bias associated with each factor.

Meta-regression for prior plausibility. The regression coefficients quantify the expected change in relative bias associated with stronger or weaker prior beliefs.

Analysts demographics. To explore whether relative or absolute bias differs by occupation, research field, experience with simulation studies, or knowledge about convenience samples, we summarize analysts’ demographic characteristics together with their mean relative or absolute bias descriptively.

7 Other

7.1 Which statistical software/packages do you plan to use?

Answer: All simulations and analyses will be conducted in R (R Core Team, 2025a) (using a current stable version at the time of the study; the exact version number will be reported in the manuscript).

The core simulation wrapper and analysis scripts rely on base R and the following packages in their CRAN versions current at the time of running the study:

- `stats` (R Core Team, 2025a) for model fitting and inferential procedures (e.g., `lm()` for linear regression);
- `ggplot2` (Wickham, 2016) for data visualization;
- `dplyr` (Wickham et al., 2023) for data manipulation;
- `tidyr` (Wickham et al., 2024) for data manipulation;
- `patchwork` (Pedersen, 2025) for combining multiple `ggplot2` figures into panel plots;
- `metafor` (Viechtbauer, 2010) for conducting the planned meta-regressions across analysts;
- `purrr` (Wickham & Henry, 2025) for functional programming tools (e.g., `map()` functions) to iterate over lists of DGPs and simulation results;
- `knitr` (Xie, 2014, 2015, 2025) for generating reproducible reports and integrating simulation output into R Markdown documents;
- `rlang` (Henry & Wickham, 2025) for tidy evaluation helpers used in programming with `dplyr/ggplot2`;
- `kableExtra` (Zhu, 2024) for LaTeX/HTML table styling;
- `ggh4x` (van den Brand, 2025) for extended faceting;
- `summarytools` (Comtois, 2025) for descriptive tables;

- `grid` (R Core Team, 2025b) for custom row and column headers in multi-panel plots;
- `RColorBrewer` (Neuwirth, 2022) for colours;
- `tidyverse` (Wickham et al., 2019) for data manipulation;
- `readr` (Wickham et al., 2026) to read in rectangular data;
- `janitor` (Firke, 2024) for data cleaning and tabulation;
- `shiny` (Chang et al., 2026) for building interactive web applications in R;
- `plotly` (Sievert, 2020) for interactive data visualisation;
- `rsconnect` (Atkins et al., 2025) to deploy shiny app.
- `forcats` (Wickham, 2023) for working with and reordering factor variables.

The DGPs will be implemented by individual analysts in separate R scripts. If additional packages are required within a particular DGP (e.g., for multi-variate data generation), these will be loaded within the corresponding DGP script and documented.

7.2 Which computational environment do you plan to use?

Answer: We will run the simulation study on a Windows 11 machine. The complete output of `sessionInfo()` will be saved and reported in the supplementary materials. Furthermore an `renv.lock` file will be uploaded to the OSF.

7.3 Which other steps will you undertake to make simulation results reproducible? (optional)

Answer: We will upload all material and code to OSF (<https://osf.io/4uwae>).

7.4 Is there anything else you want to preregister? (optional)

Answer: All materials used to recruit, instruct, and survey analysts will be preregistered, including recruitment text, introductory materials, instructions and questionnaires.

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